Embracing not avoiding generative AI to enhance scientific writing education

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The Challenge – A changing landscape

Modernizing assessment in Cell and Molecular biology

A real-world test of artificial intelligence infiltration of a university examinations system: A "Turing Test" case study

Peter Scarfe, Kelly Watcham, Alasdair Clarke, Etienne Roesch

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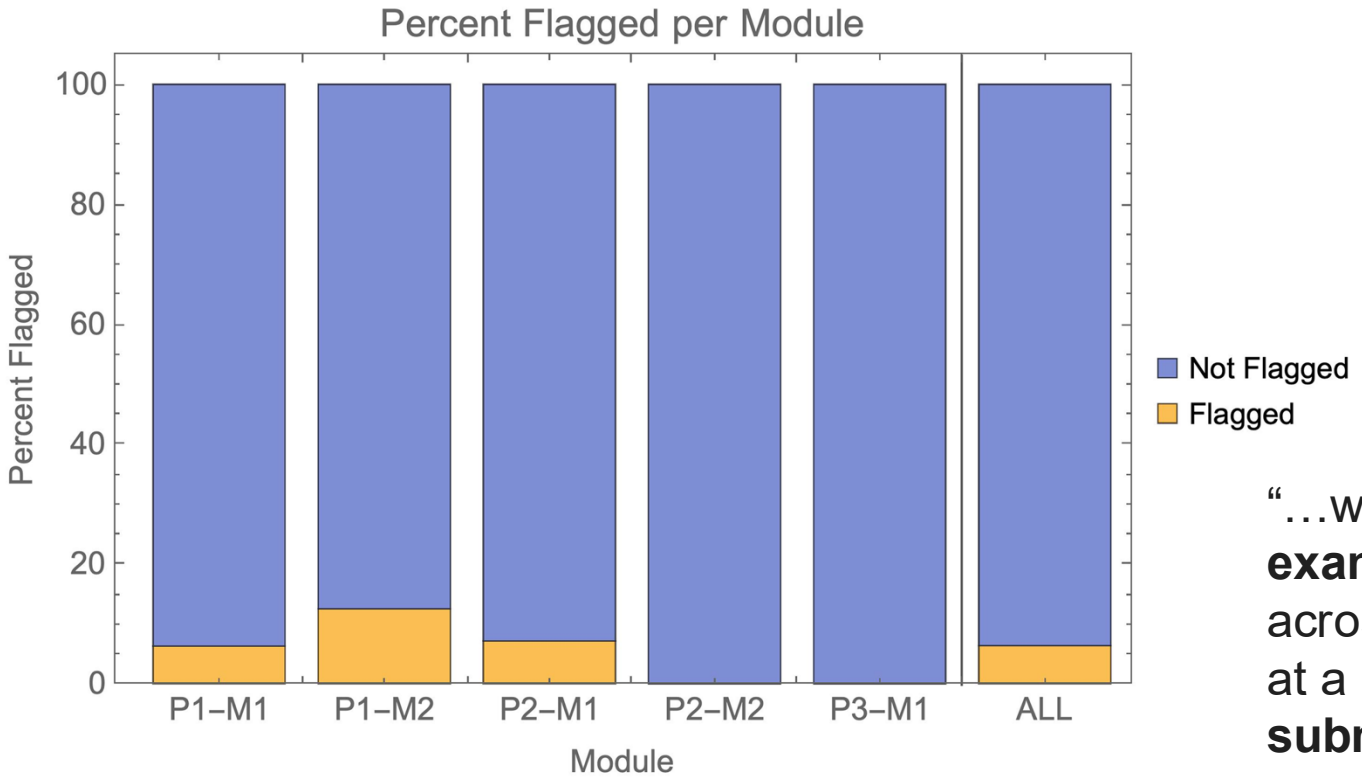


- **Generative AI is here to stay**
- **Students are using it**
- **Studies show we are unable to detect its use!**

"…we injected **100% AI written submissions into the examinations system** in five undergraduate modules, across all years of study, for a BSc degree in Psychology at a reputable UK university. We found that **94% of our AI submissions were undetected."**

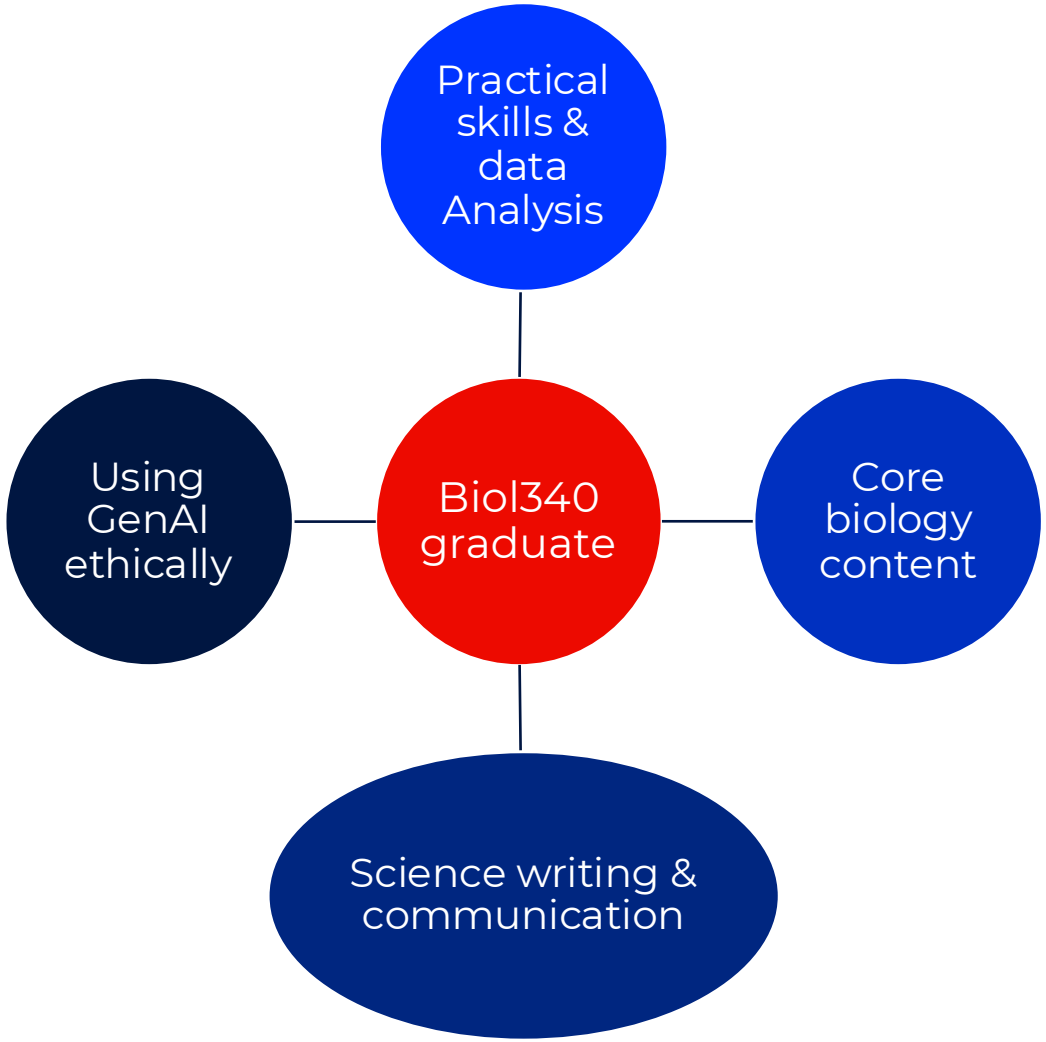
The Challenge – A changing landscape

Modernizing assessment in Cell and Molecular biology

- Computational skills including using AI are now an essential graduate skill
- We still must provide our core education in biology, practical skills and communication
- **Large cohorts constrain writing assessments** – particularly in marking
- **Biol340 requires a redesign** to provide career relevant graduate skills

The Challenge – A changing landscape

Modernizing assessment in Cell and Molecular biology

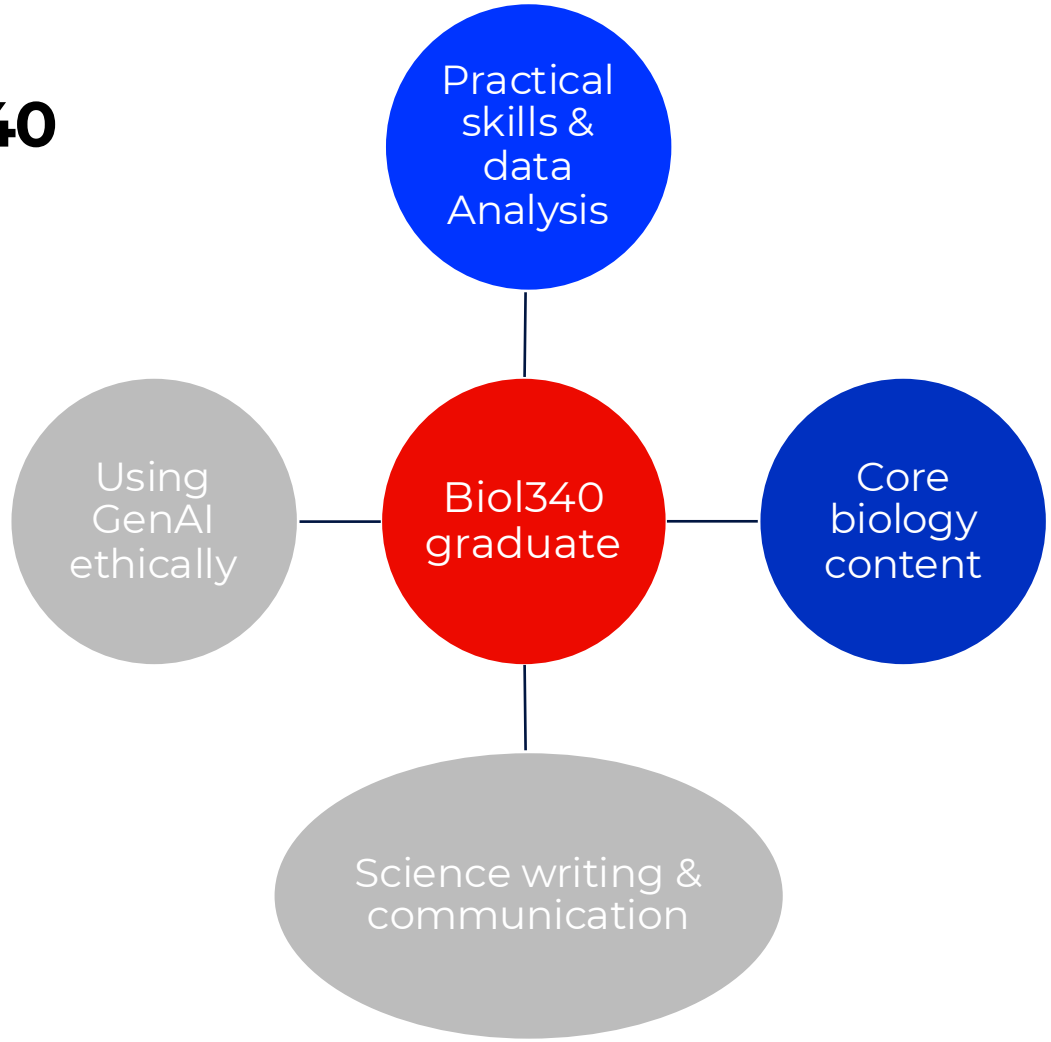


The skills of the modern graduate

The Challenge – A changing landscape

Modernizing assessment in Cell and Molecular biology

Current Biol340



The skills of the modern graduate

The Challenge – Current Biol340 Structure

Modernizing assessment in Cell and Molecular biology

Assessment

Weekly Lectures (2hr/block)

- Three theory quizzes (65% of total)

Five practical wet labs (4hr/lab)

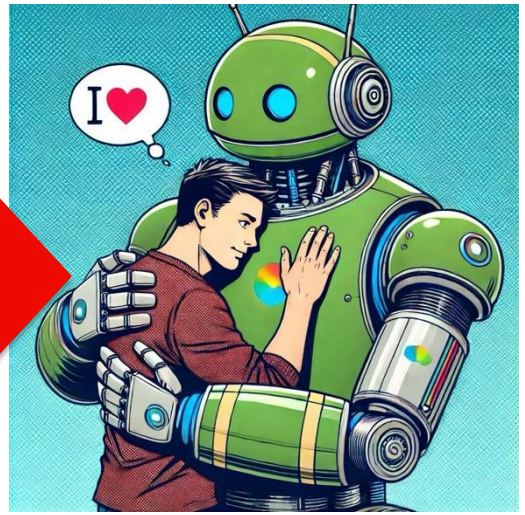
- Five pre-lab quizzes (10% of total)
- Five post-lab quizzes (25% of total)

- Class size of ~200 students most with a molecular biology background
- Student feedback that labs are too long with long period of "downtime"
- Data analysis takes place during labs but is often at the end of the classes without guided tutorial
- Current assessment model relies entirely on quizzes
- No education or assessment on scientific writing or communication
- No GenAI education

Our Aim

Redesigning Biol340

Adopt an ***embrace not avoid*** mindset for GenAI in higher education



- Update the content and assessments in Biol340 to include skills in data analysis, communication and GenAI use
- Reduce the number and length of wet labs to enable new tutorial style dry labs on data analysis and Gen AI use
- Adapt our assessments to include generative AI as a core component
- Re-introduce scientific writing assessment
- All possible through L&T project support

Redesigning Biol340 – New structure

Incorporating GenAI & computer literacy into teaching scientific writing

- Weekly Lectures (2hr/block)
  - Mid-session quiz (20% of total)
  - Final Exam (30% of total)
- Four practical wet labs (3hr/lab)
  - Four post-lab quizzes (20% of total)
  - Peer assessed scientific report (15% of total)
- Four dry labs (2hr/lab)
  - Generative AI & peer-review reflection task (15% of total)

- One wet lab removed and all shortened by 1 hr
- Four new dry labs introduced on topics:
  - Data analysis in R
  - Scientific writing
  - Peer review and feedback
  - GenAI literacy
- Two new assessments framed around a scientific report and review introduced

Redesigning Biol340 – Peer-assessed scientific writing

New Assessments

- Working in pairs to develop a piece of formal scientific writing
- Utilise data analysis skills & coding from dry pracs
- Reports are peer marked following established rubric and written feedback is given
- Peer-marking is individual
- Student sees a variety of report and can reflect on writing/communication
- Report then serves as draft for reflection assignment

Using **peer-assessment** to help inform future writing

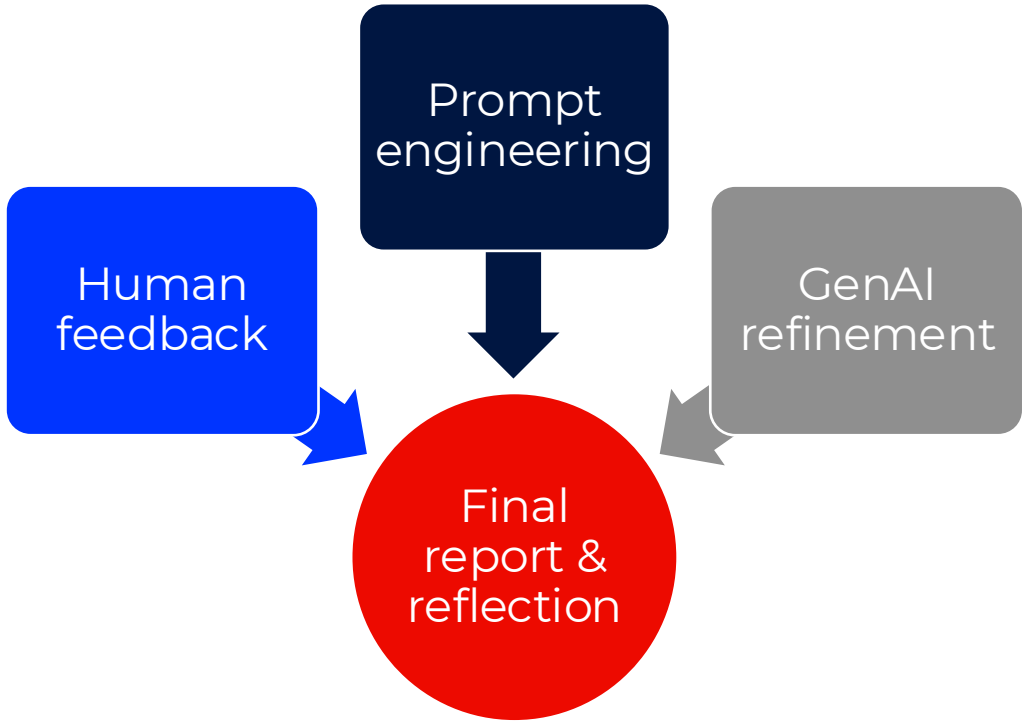
Redesigning Biol340 – Generative AI reflection piece

New Assessments

Builds of human feedback provided during peer-assessment

Using a template – students individually:

- Summarise the human feedback on their writing
- Completes a revision guided through GenAI refinement of writing using increasing levels of 'prompt complexity'
- compare their drafts pre- and post-refinement and discuss how AI changed their work
- Reflect on human vs. AI feedback and the use of AI tools for scientific writing



Learnings – Prompt engineering

Levels of complexity – getting the most from your prompts

1) “Do”

“Rewrite my discussion section to make it clearer."

2) “Evaluate”

"Evaluate the clarity of my introduction. Suggest specific improvements."

3) “Teach”

"Teach me how to improve my results section by making it more concise."

4) Model contextualisation

“You are an expert in scientific writing and an editor of a top ranked journal, give me feedback on…”

GenAI assessment example

Part 2: Revising Your Report Using AI
Instructions: Experiment with the following three levels of prompts while revising your report. Document your prompts, AI outputs, and how you applied the feedback.

Prompting Log
2.0. Contextualise the model

- **Prompt used:**
  (e.g., "You are an expert in scientific writing and an editor of a top ranked journal")

2.1. "Do for me" Prompt

- **Prompt used:**
  (e.g., "Rewrite my discussion section to make it clearer.")

- **AI Output:**
  (Paste the relevant output here)

By guiding through GenAI use we hope to:

- Inform better GenAI use
- Teach prompt engineering fundamentals
- Minimise unethical AI use (i.e. direct copy-pasting without attribution)
- Improve confidence in scientific writing & editing

The importance of contextualisation

You are a primary school student, write me an abstract for a project on mitochondrial diseases

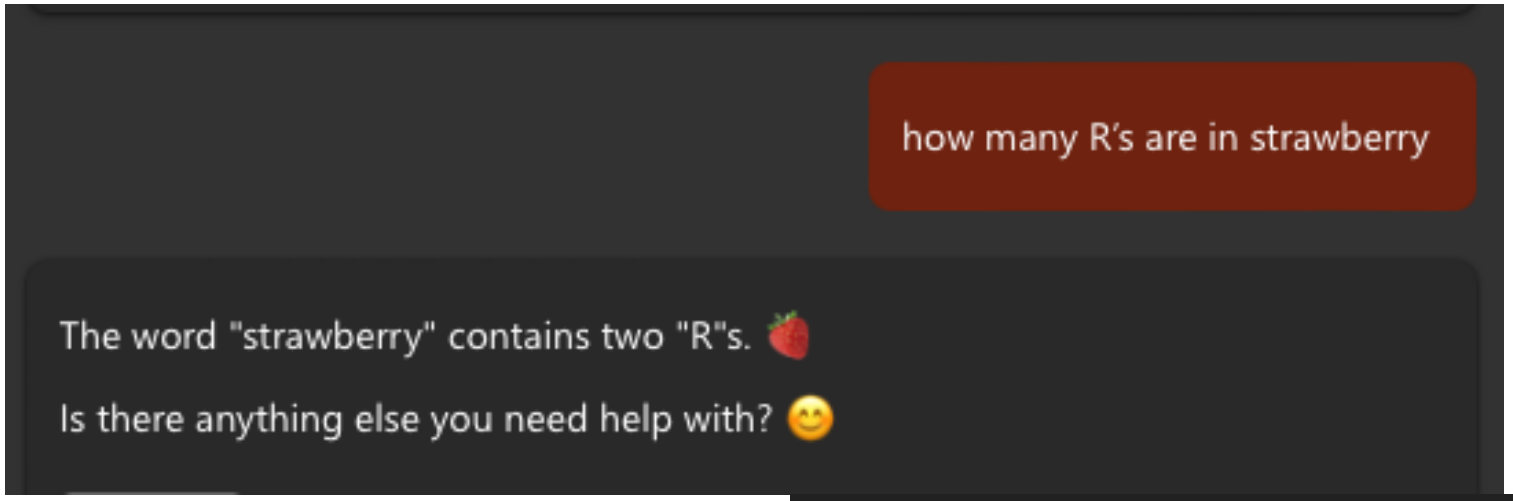
Mitochondrial diseases are when parts of our body don't work right because the **little "batteries"** inside our cells, called mitochondria, **get sick**

You are a senior lecturer and expert in biochemistry and cell and molecular biology. Write me an abstract about mitochondrial diseases

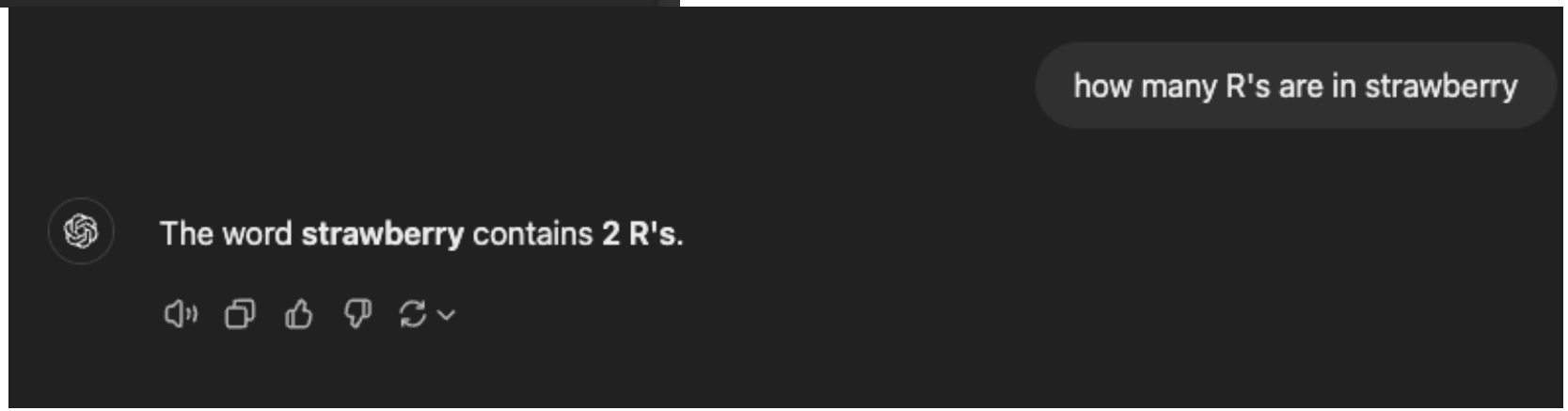
Mitochondrial diseases are a diverse group of disorders resulting from **dysfunction in** the mitochondria, **the energy-producing organelles in eukaryotic cells**. These diseases often present with multisystemic manifestations, particularly affecting high-energy-demand tissues such as the brain, muscles, and heart.

AI is a powerful tool – if used correctly

Prompt engineering and training essential!



Co-pilot



chatGPT-4

AI is a powerful tool – if used correctly

Prompt engineering and training essential!

chatGPT-4

You are an educator in linguistics with expertise in writing. Evaluate how many R's there are in strawberry

As a linguist and writer, let's break this down systematically:

The word **strawberry** contains the following sequence of letters: **s-t-r-a-w-b-e-r-r-y**. Now, let's count the **R's**:

1. The first **R** appears in the third position (**st→r**).
2. The second **R** appears in the eighth position (**be→r**).
3. The third **R** appears in the ninth position (**r→y**).

Thus, **strawberry** contains **3 R's**.

This method ensures accuracy by verifying each occurrence systematically.

Learnings –other takeaways

Embrace not avoid

- Curriculum restructure requires careful consideration of the skills needed by biology graduates
- Integration of data analysis and GenAI has been completed using a step-wise multiple assessment approach
- By encouraging AI use we are equipping students with skills to use this technology more effectively
- Our reflection assessment actively asks for output from GenAI – which discourages cheating & plagiarism
- Through our approach we can reintroduce scientific writing education in an innovative way

Next steps

Rolling out these changes & getting feedback

- The redesigned BIOL340 will be rolled out in 2025
- Plans are in place to monitor and refine the curriculum based on student feedback
- We are currently developing feedback mechanisms to allow us to assess how students engage with the new content and computational tools
- Teaching staff feedback will also be gathered
- Further refine subject for 2026
- Prepare for further integration with Co-Pilot as it is rolled out & championed by Microsoft

Thanks to our teaching team in Biol340 & L&T

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